2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI), 8-9 July 2021, Rajshahi, Bangladesh

Text and Voice to Braille Translator for Blind People

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***Abstract*— Braille is the main way to teach blind people. Blind readers can read using this Braille system with the help of their fingers. Much work has been done to modernize the blind education system, but it is still very expensive and complex. This paper proposed a cost-effective and user-friendly braille device based on simple technology. This paper presents the design, implementation, and working principle of a new concept of a Braille translating device which can convert text and voice to braille signs for blind students. It can convert letters from different languages to Braille symbols according to the "Voice" and "Text" commands given by the user. It can give an output like a traditional Braille sign with a voice, for confirmation of the sign displayed in the cell. This device consists of only one refreshable braille cell which can show those braille signs. Along with blind students, educators can take advantage of this device to teach blind students. Educators can operate this device with the help of smart phones and computers using Bluetooth and a USB cable. Educators can connect more than 30 braille translating devices with their computers and teach 30 students at a time. After user test and result analysis of this proposed device, observed that this system's portability, simplicity, cost efficiency, and user-friendly interface put this ahead of other devices. The results suggest that the given approach is a superior option for implementing refresh-able braille devices more simply and cost-effectively.**

Keywords*— *assistive device, braille, servo motor, braille cell, refreshable braille cell.

I. Introduction

The World Health Organization (WHO) released a statement on the number of blind people living in our world in 2011 (GLOBAL DATA ON VISUAL IMPAIRMENTS 2010) [1]. Globally, the number of people of all ages visually impaired is estimated to be 285 million, of whom 39 million are blind and many of them have blind students and 246 million have low vision. About 90% of them live in low-income developing countries. Because of this, educating them or providing services has become a serious challenge for governments, NGOs, health organizations, and society. This is because the methods by which blind students are currently receiving an education are very expensive. As a result, even if they want to, they will be unable to obtain an education. At present, they receive an education in Braille. The scientist "Louis Braille" [2] invented this method in 1824, using the Braille method "Fig. 1" for the visually impaired to read and write.

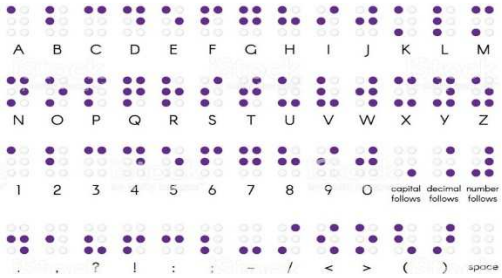


Fig. 1. Basic braille dot representation.

Braille is an arrangement of raised dots that can be read with the fingers by people who are visually impaired or who have extremely low vision [3]. Braille is not a language. Rather, it is a code that is created through a combination of some dots. Currently available braille systems book, booklet, and modern device. One of the modern devices is the Braille display of 12 to 80 cells. But the cost of these books and devices is expensive and everyone can't get an education in this way. Because the cost of "Braille printing" [4] and the price of Braille devices are skyrocketing [5]. For this reason, we have started working with the braille method of this new technology. This device especially focused on braille learning, reading, and braille teaching systems. The braille cell of the device is created at a very low cost and with a very simple mechanism using only two micro servo motors with two rotating wheels. It aims to make it easier for the blind student to learn the braille sign and read the "Text" file using the device. It is very small in size and portable. As a result, we can move with this device very easily. It can take input in several different languages as a voice or a text command and translate to braille sign. And it is designed in such a way that the braille dot can be easily touched by the fingers of the hand. It takes voice input through a mobile application to convert voice to "braille sign". The mobile application then uses the "speech to text engine" to convert voice to text. The text is subsequently sent to the device as a user input for braille conversion.

978-1-6654-3843-8/21/$31.00 ©2021 IEEE

Since the mid-1970s, piezoelectric and magnetic technologies have been developed as commercial pin actuators for refreshable Braille displays, piezoelectric technology [6]was used in braille cells, since the complexity and limitations of the magnetic actuator between these two technologies were possible to avoid in piezoelectric actuators. Subsequently, some new technologies were developed by researching the "Braille Cell". Some notable technologies are: electrothermal actuator [7], Braille Automatic Recognition Method [8], Microvalve technology [9].

Much work is being done worldwide on the modernization of the blind education system. In its continuation, various types of Braille devices have been invented. However, these devices are very expensive and rely on complex technology. Currently, a lot of work is being done around Braille devices to make this education system easier and more accessible. A M Muntasir Rahman proposed a method of refreshable braille display using micro servos in [10], which requires 6 servo motors to control the 6 dots of a cell and this method did not maintain the standard size of a braille cell and dot. In [11], A Single-Cell Electromagnetic Refreshable Braille Display was proposed by Gemma Carolina Bettelani. in this method need 6 "Linear Solenoid Actuator" for controlling the 6 dots of a braille cell. The power consumption rate of each actuator is 1W and every actuator generates a certain amount of heat, but the mechanisms of these devices are very complex and expensive. Xiaosong Wu [12] developed a portable pneumatically actuated refreshable braille cell. In this system it needs high pressure air flow for controlling dot actuation. These devices have no voice recognition system and these devices can not give mechanical and voice output at a time.

II. Methodology

Now the question may arise, how to read text files or books using one Braille cell? Usually, a refreshable braille display consists of 12 to 80 cells for reading text file or book. As the braille reader fell one cell at a time and go for next cell. We decide to give this feeling of reading using one cell only instead of multiple cells. Like the finger move over the cells, here the reader has to rest his/her finger over the cell and the cell is refreshed then shows Braille's meaningful signs one by one. As a result, it is possible to read with a single cell. The steps outlined in this device below are shown in Fig. 2 through a flowchart.

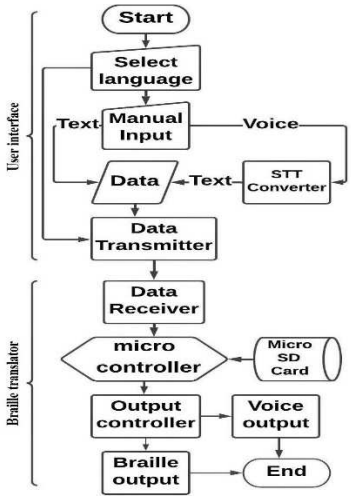


Fig. 2. Workflow diagram of text and voice to braille translating method.

A. Select Language

People all over the world are using different kinds of languages to communicate with each other. These languages vary from region to region. This device consists of a language pack. This language pack consists of a different kind of languages for translating text to braille. So, at first we need to select a specific language to send to the braille translator

B. Take Input

For this device, a user interface software for taking input from users is used. As an input it can take voice and text from user with the help of mobile or computer. If the input is voice, then the voice has to be converted to text via a Speech to Text converter. And this text file is our desired data.

C. Data Transmitter and Receiver

For communication between user interface (software) and Braille Translator, Bluetooth (as wireless communication) in "Fig. 3" or Serial USB connection has been used with a bandwidth rate of 9600 bps. As we are using a 8 bit microcontroller so it can't read a whole UNICODE [13] character at a time so every characters has been divided into some reasonable section and send to the translator packet by packet.

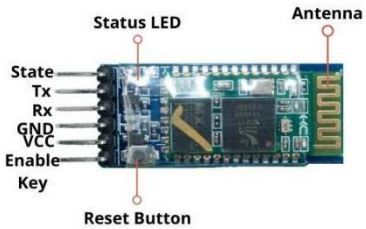


Fig. 3. Bluetooth module.

D. Microcontroller

We know that microcontroller work as a brain of a device. it controls the whole device. After taking input data from the user, it observes and analyses them then takes the decision about what will be output. For giving the output as a braille sign it needs to control two motors. According to the user input, the microcontroller controls the motor's rotation and the user gets desired output as a braille sign.

E. Output

It is able to give two types of output, the first one is the braille sign and the other one is a voice output. As a result, blind student can read using their finger and they hear a voice output of what they read.

F. Storage

This device has an SD card module, it can store text files and books. So that the user can read the text file from the SD card.

III. Braille Cell Design and Dot Combination

Braille consists of patterns of raised dots arranged in cells which is up to 6 dots in 3 rows and 2 columns configuration. In this configuration every column consists maximum 3 dots or every row consist maximum 2 dots which has given on "Fig. 4". With these 6 dots of a cell, more than 64 braille signs can be made, Because 2^6=64. Each cell represents a braille letter, numeral or punctuation mark.

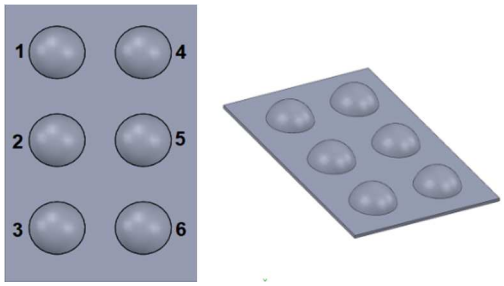


Fig. 4. Top view and dot numbering of braille cell.

In Fig. 5 its visible that coloum to coloum distance is 2.3 mm and height is (0.6-0.9) mm. Row to row distance is 2.5mm and base diameter is (1.5-1.4) mm.

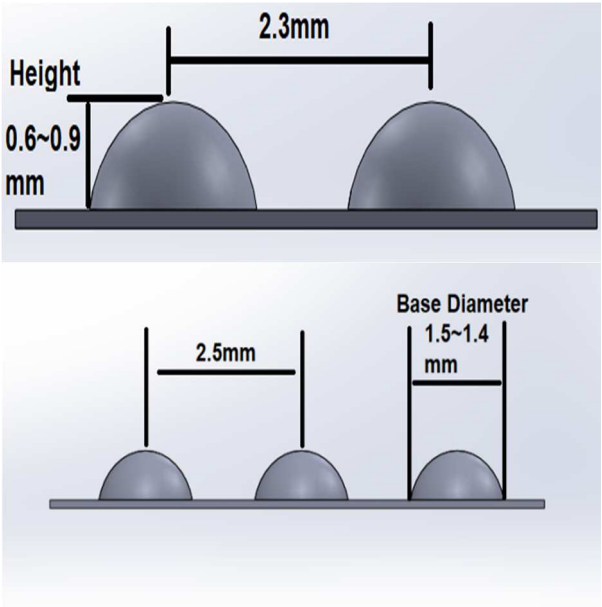


Fig. 5. Parameters for braille dots.

A column of Braille cells can contain a maximum of three dots. And these three dots can create different possible combinations among themselves. We have illustrated all the possible combinations in a column. These combinations are placed on each rotating wheel of the Braille cell.

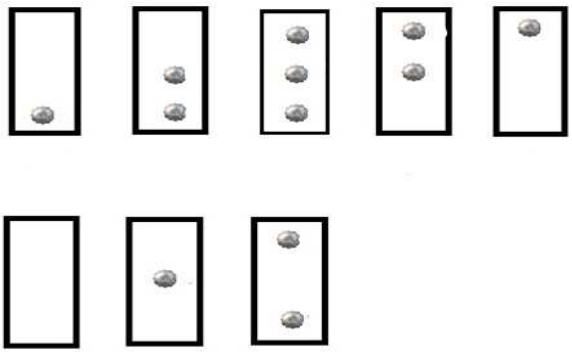


Fig. 6. Braille cell dot combination.

There are two rotating wheels for making this braille cell. Each wheel is connected vertically to a servo motor (9g) and a microcontroller has been used on the device to control these motors. To show the braille sign, dots are placed on the wheels, which can be felt with the fingers of the hand. The dots are positioned at certain distances on the wheels, so that the wheels change their angle according to a command to show a meaningful braille sign. An individual sign indicates an individual character. This device takes some processing time to refresh the braille cell for character to another character. The duration of changing characters depends fully on the user and how fast the operator can read it. From a test run it was seen that a beginner can read three characters per second whereas for a professional it takes six characters.

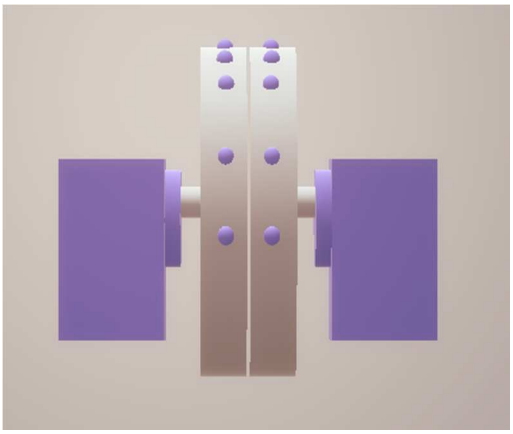


Fig. 7. Braille cell

The servo can rotate 180 degrees (90 degrees in each direction) and operates in the same way as regular servos but in a smaller size.

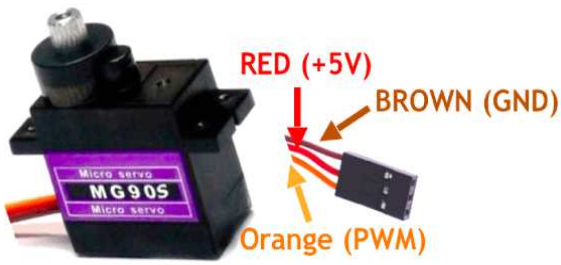


Fig. 8. Servo motor.

To operate these servos, we can use any servo code, hardware, or library. Using by PWM signal it can be controlled, which signal provided from the microcontroller for controlling this motor.

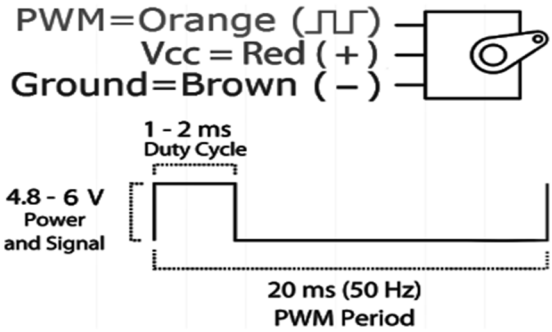


Fig. 9. Motor duty cycle and PWM period.

Position ‘0’ (1.5 ms pulse) is middle, ‘90’ (~2 ms pulse) is all the way to the right, ‘90’ (~1 ms pulse) is all the way to the left.

TABLE I. MOTOR CONFIGURATION

Modulation	Analog
Torque	4.8V: 30.60 oz-in (2.20 kg cm) 6.0V: 34.70 oz-in (2.50 kg-cm)
Speed	4.8V: 0.11 sec/60° 6.0V: 0.10 sec/60°
Weight	0.49 oz (14.0 g)
Dimensions	Length:0.91 in (23.1 mm) Width:0.48 in (12.2 mm) Height:1.14 in (29.0 mm)
Motor Type	Mini
Gear Type	Metal
Rotational Range	180°
Pulse Cycle	20 ms
Pulse Width	400-2400 µs

IV. SIMULATION & PRACTICAL EXPERIMENT

A microcontroller board named "Arduino nano" has been used to control the entire process of this device. This can actually be compared to the brain of the device. The board has a microcontroller (ATmega328P – 8-bit AVR family microcontroller) with 2KB SRAM, 1KB EEPROM, runs on 16MHz, 32kb Flash Memory, and Operating Voltage 5V. A DC 12V 3A battery is used as a power source with a voltage regulator to gain 5V operating voltage for the microcontroller. The HC-05 Bluetooth module has four pins, namely: VCC, GND, TX, RX. The first two pins are for power supply and the rests are for communication with the microcontroller and mobile device. It includes an SD card slot for storing text files and books. In order for the user to be able to read the text file from the SD card. The SD card module has 6 pins, they are VCC, GND, CS, SCK, MISO, MOSI. The first two pins are for power supply and another 4 are for data transfer between the SD card and the microcontroller. The servo motor has three pins, the pins are: Signal, VCC, GND. Here, VCC and GND pins are for power supply and the signal pin connected to microcontroller to take instructions.

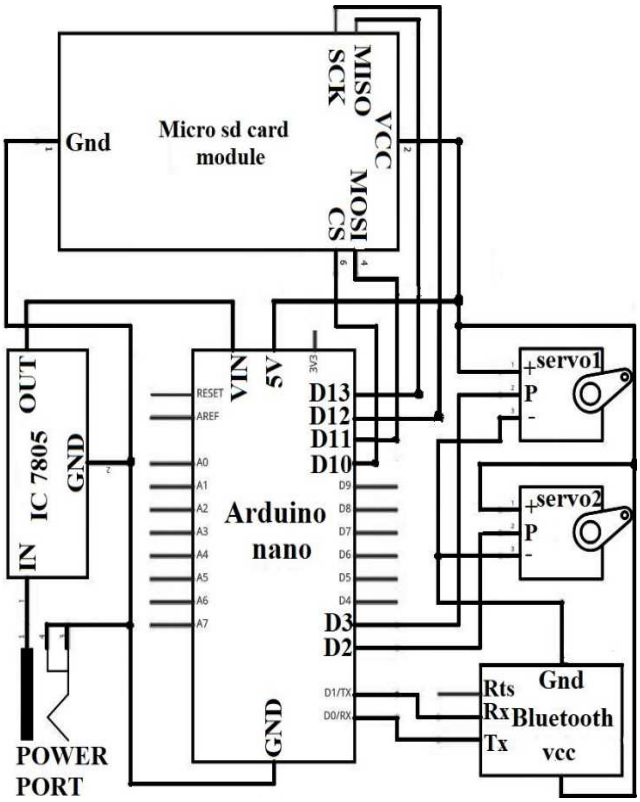


Fig. 10. Circuit diagram of braille system.

A. Desktop Application

There is a desktop(windows) application to manage this device from computer. By using this application, a teacher can teach students with a great efficiency. A teacher can teach several students at a time using this application while a student can use it for self-learning with an extra feature of voice output.

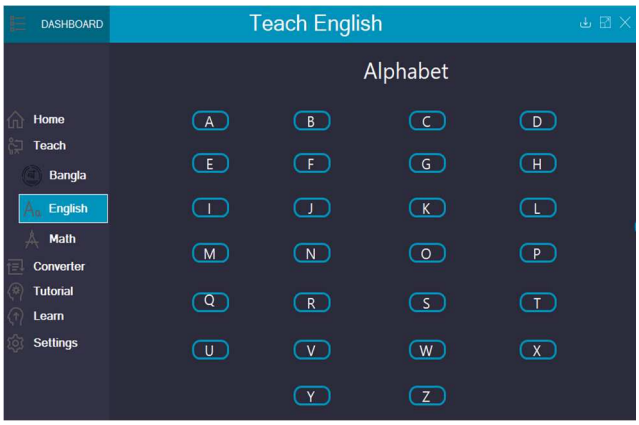


Fig. 11. User interface of desktop application.

B. Mobile Application

With similar features of desktop application mobile application is mainly designed for using with a single device. Also, it gives the extra features of voice to braille output in many different languages.

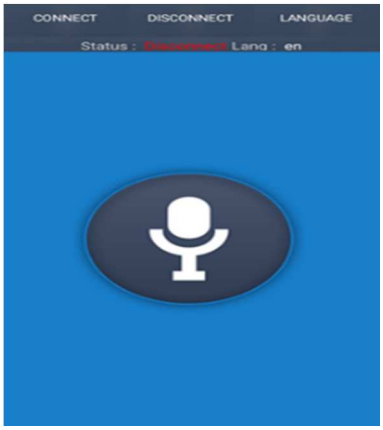


Fig. 12. User interface of mobile application.

"Fig. 13" shows, the user touches the refreshable Braille cell with the finger of his hand then gives a voice input using the mobile application. The device displays the Braille sign according to the voice input and gives a voice output according to the accompanying Braille sign. The user can feel this braille sign using his finger and hear the voice for a better understanding.

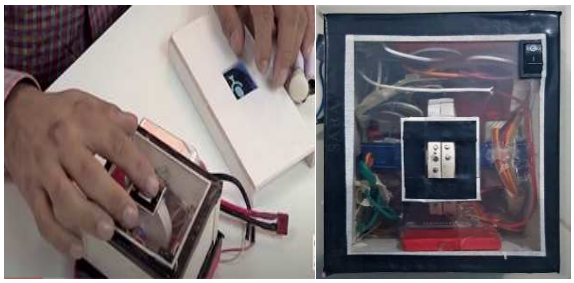


Fig. 13. Reading experiment and braille cell of our device .

"Fig. 14" shows, this device has been tested by a user. At the end of the test, he talked about different aspects of the device. According to him, the device is very good at converting text and voice to Braille sign which he can understand by using his fingers without any problem. He believes that by using this low-cost device, blind students will be able to learn digitally and it's quite easy to operate compared to other expensive devices.

Fig. 14. Visually impaired person test the device.

V. PERFORMANCE

To test the understandability of this device we've tested it with different persons and got a 100%-character detection accuracy. Also, it can be used as a braille display it can automatically change cell orientation according to the next character without any user interaction. And there we have got a 3 character per second for a learner or children, 6 character per second for an expert reader. This device has a response time of only 5 milliseconds.

VI. CALCULATION & COST ANALYSIS

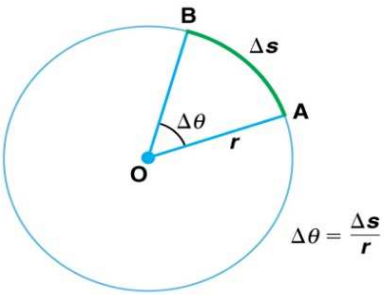
For changing its dot position, it follows Angular Displacement law. The rotating wheels of this device moves in a circular path.

Here,

δs is displacement of 'A' to 'B'

'R' is the radius of rotating wheel, $\delta\theta$ is the angular displacement which is got from servo motor.

$$\delta\theta = \frac{\delta s}{r} \tag{1}$$



For servo (mg90s) motor, rotational range is 180°. For this reason, useable angular displacement must be less than 180°

$\delta\theta < 180°$

From figure 6, we can see that each rotating wheel needs to contain 8 dot combinations. For each dot combination it needs 7mm as a length of the arc of the rotating wheel.

So that minimum length of the arc of the rotating wheel is

(8 * 7 mm) = 56mm.

Now we can find out the minimum radius of the rotating wheel for contain 8 dot combinations.

Here we have tried to build something better at a low cost. So that visually impaired students can get quality education within their financial ability.

TABLE II. COST ANALYSIS

NO.	Parts name	Price
1	Microcontroller	3USD
2	Servo motor	(1.2*2) USD
3	HC-05 Bluetooth module	2.1 USD
4	3D printed body	2 USD
5	Battery	1.5 USD
6	PCB	1.2 USD
7	Connecting wire	0.5 USD
8	SD card module	1 USD

VII. CONCLUSION

This refreshable braille translator can translate from a variety of languages to braille signs, allowing blind students to learn and read in a digital format. It is possible to learn the Braille method in a simple and modern manner with the aid of this device at a very low cost. As a result, it will be able to play an important role in the blind education system. Following the consumer test, we discovered that visually disabled people can use this single cell braille translator to read and learn braille signs. As this device cannot take direct command from users, it requires a smartphone or a computer to take command from humans. We plan to produce a standalone version of this device in the future. Which will be able to function without the use of a computer or a smartphone. For the user's comfort, we will upgrade this device with new technology. We have received a lot of positive feedback from the users, who were mainly students and teachers, and decided to put the device on the market.

ACKNOWLEDGMENT

We'd like to thank "Dr. Umma Hanny", "Dr. Fakir Sharif Hossain", and "Shah Md. Salimullah" for their support and advice. We are extremely grateful to "Md. Nahidul Alam" since we acquired the knowledge of robotics and paper writing from him and we got maximum support from him in every instance. Thanks also to "Taiyeb Hasan Sakib" ,"Md. Jahid Hasan" and " Iftekher Alam Rahat " without whom we would not have been able to write this paper.

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